NPTEL Online Certification Course
Applied Accelerated Artificial Architecture
Assignment 10 : Detailed Solution
Indian Institute of Technology Guwahati

Q1. Which operation is considered the most expensive in a distributed data processing pipeline because it involves regrouping and moving data across workers?

- A. Evaluation
- B. Shuffle
- C. Partitioning
- D. Lineage

Correct Answer: Shuffle
Explanation: Shuffling redistributes data across different partitions, which heavily utilizes network bandwidth, disk I/O, and data serialization. This massive movement of data between worker nodes naturally makes it the most resource-intensive operation in distributed computing.

Q2. Which of the following best defines the concept of "data lineage" within a distributed data engineering architecture?

- A. A dynamic layout mechanism that optimizes physical disk storage distribution.
- B. A deterministic profile that calculates network latency between operational workers.
- C. A record of data derivation that facilitates state reconstruction in the event of failure.
- D. An immediate evaluation strategy that executes instruction blocks line by line.

Correct Answer: A record of data derivation that facilitates state reconstruction in the event of failure.
Explanation: Data lineage tracks the exact sequence of transformations applied to a base dataset, creating a logical execution plan. If a node fails, the system uses this lineage graph to recompute only the lost partitions from the original data instead of restarting the entire pipeline.

Q3. How does a tool like Spark or Dask handle instruction execution differently compared to Pandas?

- A. It relies heavily on strict static allocation
- B. It executes each individual line immediately

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- C. It uses a highly centralized row range lookup
- D. It employs an optimization via lazy evaluation

Correct Answer: It employs an optimization via lazy evaluation
Explanation: Unlike Pandas, which executes operations eagerly line-by-line, frameworks like Spark build a directed acyclic graph (DAG) of transformations. Execution only begins when an action is called, allowing the engine to optimize the entire query plan for maximum efficiency.

Q4. In distributed data systems, how does a scheduler notice that a worker has unexpectedly failed mid-group?

- A. It detects that the worker stopped heartbeating
- B. It scans the primary key for missing data entries
- C. It identifies that a local task graph has ended
- D. It encounters an automatically forced row range

Correct Answer: It detects that the worker stopped heartbeating
Explanation: Worker nodes periodically send quick "heartbeat" signals over the network to the central cluster manager to confirm they are active. If the scheduler stops receiving these heartbeats within a specific timeout window, it assumes the worker has crashed and reassigns its tasks.

Q5. Which partitioning scheme is ideal for a clickstream group by user ID to achieve zero shuffle?

- A. Hashing
- B. Salting
- C. Row range
- D. Range date

Correct Answer: Hashing
Explanation: By hashing the user ID, all records belonging to the same user will consistently map to the exact same partition. When a "group by" is performed on this ID, the data is already localized on the correct node, eliminating the need to shuffle data across the network.

Q6. Which framework is nearly identical to Pandas and best suited for tabular ETL ranging from 10 GB to 10 terabytes?

- A. A stateless function
- B. A cloud-managed storage pod
- C. A data partitioning scheduler
- D. A stateful Python object

Correct Answer: A stateful Python object
Explanation: While phrasing is unconventional, distributed architectures for tabular ETL

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(like Ray or Dask) rely on stateful Python objects (Actors) to scale processing. These tools maintain state across nodes to handle datasets that exceed single-machine memory constraints.

Q7. Which deployment model allows for dynamic autoscaling of pods when working with Dask?

- A. Local cluster
- B. Kubernetes cluster
- C. Cloud-managed service
- D. Single core machine

Correct Answer: Kubernetes cluster
Explanation: Kubernetes natively supports dynamic resource allocation, allowing distributed frameworks like Dask to seamlessly spin up or tear down worker pods based on the real-time workload. This elastic autoscaling ensures optimal resource utilization and cost efficiency.

Q8. What is a common mistake when managing frameworks, which leads to poor GPU utilization?

- A. Using Ray tune for model serving
- B. Choosing Dask for GPU inference
- C. Deploying Ray tasks on Kubernetes
- D. Applying Pandas APIs to Dask arrays

Correct Answer: Choosing Dask for GPU inference
Explanation: Dask is primarily designed for distributed task scheduling and data processing, not for managing the strict memory complexities of GPU model inference. Utilizing specialized serving frameworks is necessary to achieve high throughput and avoid severe GPU bottlenecks.

Q9. What is the purpose of "salting" in the context of handling data skew?

- A. It deletes duplicate rows from the partition.
- B. It caches data frames directly into the CPU memory.
- C. It adds a random value to a skewed key to spread rows.
- D. It converts a Python user-defined function into SQL.

Correct Answer: It adds a random value to a skewed key to spread rows.
Explanation: Data skew occurs when one key has vastly more records than others, overloading a single partition. Salting adds a randomized prefix or suffix to this problematic key, artificially breaking the massive group into smaller sub-keys that distribute evenly across nodes.

Q10. For which size of data sets will GPUs not provide any processing benefit when using CUDF?

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- A. Less than 100k rows
- B. Around 10 million rows
- C. More than 50 million rows
- D. Petabyte-scale data sets

Correct Answer: Less than 100k rows
Explanation: For very small datasets, the overhead time spent transferring data from the CPU to the GPU's memory (the PCIe bottleneck) takes longer than the actual computation. CPUs are already fast enough to process sub-100k rows instantly, making GPU acceleration counterproductive.

Q11. Which component in Spark automatically rewrites queries to optimize performance and speed?

- A. Executor
- B. Delta Lake
- C. Cluster Manager
- D. Catalyst Optimizer

Correct Answer: Catalyst Optimizer
Explanation: The Catalyst Optimizer is Spark's advanced query planning engine that parses SQL and DataFrame operations into an optimized physical execution plan. It applies smart rules like filter pushdown and column pruning to heavily reduce data scanning times.

Q12. What common mistake in Spark causes the driver to run out of memory and crash?

- A. Calling .collect() on a huge data set
- B. Utilizing Adaptive Query Execution
- C. Writing data directly to the storage
- D. Processing data sets with small keys

Correct Answer: Calling .collect() on a huge data set
Explanation: The `.collect()` operation forces all distributed worker nodes to send their entire partitioned data back to a single driver node. If the resulting dataset exceeds the driver node's allocated memory capacity, it throws an OutOfMemory (OOM) error and crashes.

Q13. Which of the following is defined as a discrete, timestamped record of something that actually happened in a system?

- A. Metric
- B. Log
- C. Trace
- D. SLO

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Correct Answer: Log
Explanation: A log is an immutable text record detailing a specific event or transaction that occurred within an application at an exact moment in time. Unlike aggregated metrics, logs provide granular context regarding errors and systemic behaviors for debugging.

Q14. In the provided ETL troubleshooting scenario, what was diagnosed as the root cause for the runtime jumping from 40 minutes to 3 hours?

- A. Disk storage failure
- B. Data skew from a new bot account
- C. Network packet loss
- D. Wrong schema definition

Correct Answer: Data skew from a new bot account
Explanation: A sudden, dramatic increase in ETL runtime is frequently caused by data skew, where a massive influx of anomalous records (like spam bot activity) maps to a single partition. This creates a "straggler task" where one node bottlenecks the entire pipeline.

Q15. When investigating a data pipeline bottleneck, in what specific order should you analyze the three complimentary signals?

- A. Traces, Metrics, Logs
- B. Metrics, Traces, Logs
- C. Logs, Metrics, Traces
- D. Traces, Logs, Metrics

Correct Answer: Logs, Metrics, Traces
Explanation: Analyzing logs first allows engineers to identify exact error messages or immediate anomalies at the timestamp of the bottleneck. Once isolated, metrics can quantify the system's overall resource health, and traces map the request's path to find the specific latency source.

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